

Lomax *et al.*

- 37 Pezaro C, Mukherji D, Tunariu N, Cassidy AM, Omlin A, Bianchini D *et al.* Sarcopenia and change in body composition following maximal androgen suppression with abiraterone in men with castration-resistant prostate cancer. *Br J Cancer* 2013; **109**: 325–31.
- 38 Attard G, Reid AH, Auchus RJ, Hughes BA, Cassidy AM, Thompson E *et al.* Clinical and biochemical consequences of CYP17A1 inhibition with abiraterone given with and without exogenous glucocorticoids in castrate men with advanced prostate cancer. *J Clin Endocrinol Metab* 2012; **97**: 507–16.
- 39 Gardner JR, Livingston PM, Fraser SF. Effects of exercise on treatment-related adverse effects for patients with prostate cancer receiving androgen-deprivation therapy: a systematic review. *J Clin Oncol* 2014; **32**: 335–46.
- 40 Keogh JW, MacLeod RD. Body composition, physical fitness, functional performance, quality of life, and fatigue benefits of exercise for prostate cancer patients: a systematic review. *J Pain Symptom Manage* 2012; **43**: 96–110.
- 41 Galvão DA, Taaffe DR, Spry N, Newton RU. Physical activity and genitourinary cancer survivorship. *Recent Results Cancer Res* 2011; **186**: 217–36.
- 42 Livingston PM, Craike M, Salmon J, Courneya K, Fraser S, Gaskin C *et al.*, the ENGAGE Uro-Oncology Clinicians' Group. The efficacy of a clinician referral and 12 week physical activity program among men living with prostate cancer: a multicentre cluster randomised controlled trial [ENGAGE]. *Cancer* 2016 (in press).
- 43 Basaria S, Bhasin S. Targeting the skeletal muscle-metabolism axis in prostate-cancer therapy. *N Engl J Med* 2012; **367**: 965–7.

CLINICAL PERSPECTIVES

Autoimmune encephalitis

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Key words

autoimmune encephalitis, limbic encephalitis, neuronal surface antigen antibody, anti-NMDAR, anti-VGKC.

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Received 20 July 2015; accepted 19 November 2015.

doi:10.1111/imj.12974

Abstract

Over the past decade, the clinical spectrum of autoimmune encephalitis has expanded with the emergence of several new clinicopathological entities. In particular, autoimmune encephalitis has recently been described in association with antibodies to surface receptors and ion channels on neurological tissues. Greater clinician awareness has resulted in autoimmune encephalitis being increasingly recognised in patients with unexplained neurological and psychiatric symptoms and signs. The clinical spectrum of presentations, as well as our understanding of disease mechanisms and treatment regimens, is rapidly developing. An understanding of these conditions is important to all subspecialties of Internal Medicine, including neurology and clinical immunology, psychiatry, intensive care and rehabilitation medicine. This review provides a contemporary overview of the aetiology, investigations and treatment of the most recently described autoimmune encephalitides.

Introduction

Limbic encephalitis was first described in 1968 as a constellation of symptoms affecting cognition and behaviour, frequently associated with seizures.¹ Originally, most limbic encephalitides reported were paraneoplastic, associated with lung, breast, ovarian or testicular

malignancies and had poor prognoses.² Paraneoplastic antibodies were found to act against intracellular proteins, the best characterised of which are anti-Hu/ANNA-1, anti-Ri/ANNA-2, anti-CV2/CRMP5 and anti-Ma2/Ta.²

More recently, autoantibodies against the neuronal cell surface or synapse have been described (Table 1). These antibodies are likely directly pathogenic, interfering with cell surface proteins. The most commonly encountered antibodies of this type are the anti-N-

Funding: None.
Conflict of interest: None.

Table 1 Autoimmune encephalitis syndromes associated with antibodies against the neuronal cell surface and synaptic antigens

Antigen	Clinical features	Age range (median) years	Sex ratio (F:M)	Tumour associations and frequency
NMDA receptor (NR1 subunit) ^{3,6}	Prodromal syndrome Psychiatric manifestations, seizures, amnesia, movement disorder, autonomic instability	0.6–85 (21)	4:1	Age dependant, ovarian teratoma, 10–50%
LG1 ^{20,21}	Limbic encephalitis, seizures	30–80 (60)	1:2	Thymoma, thyroid, lung, renal cell carcinoma reported <10% for all tumours combined
CASPR2 ^{20,28}	Morvan's syndrome, neuromyotonia, limbic encephalitis (less common)	46–77 (60)	1:4	Thymoma, 0–40%
AMPA receptor (GluR 1/2) ³²	Limbic encephalitis, psychiatric syndromes	38–87 (60)	9:1	Lung, breast, thymoma, thymic carcinoma Up to 70% for all tumours combined
GABA(b) receptor ³⁰	Limbic encephalitis, seizures	16–77 (62)	2:3	SCLC 50% (predominantly older patients)
GABA(a) receptor ³¹	Refractory seizures, status epilepticus	3–63 (22)	1:5	Thymoma in one of six cases
Glycine receptor α 1 ⁴¹	PERM, limbic encephalitis, stiff person syndrome	1–75 (49)	Adults 6:5 Children 4:0	Thymoma (most commonly), lymphoma, breast Up to 10% for all tumours combined
mGluR5 ^{42,43}	Limbic encephalitis (Ophelia syndrome), myoclonus	15–46	1:1	Hodgkin lymphoma 3/3 cases reported
DPPX ^{44,45}	Encephalitis, CNS hyperexcitability, agitation, hyperekplexia, movement disorders, seizures, PERM	15–76	1:2	No tumours described
DR2 ⁴⁶	Basal ganglia encephalitis, movement disorders, psychiatric manifestations	1.6–15 (5.5)	1:1	No tumours described

AMPA, α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid; CASPR2, contactin-associated protein-like 2; CNS, central nervous system; DPPX, dipeptidyl-peptidase-like protein-6; DR2, dopamine 2 receptor; F, female; GABA, gamma-aminobutyric acid; GluR1/2, glutamate receptors 1 and 2; GlyR, glycine α receptor; LGI1, leucine-rich, glioma-inactivated 1; M, male; mGluR5, metabotropic glutamate receptor 5; NMDA, N-methyl-D-aspartic acid; PERM, progressive encephalomyelitis with rigidity and myoclonus; SCLC, small-cell lung cancer.

methyl-D-aspartic acid (anti-NMDA) receptor antibody and the anti-leucine-rich, glioma-inactivated 1 (anti-LGI1) (voltage-gated potassium channel (VGKC)) antibody. It is probable that many cases previously labelled as 'idiopathic' limbic encephalitis or 'steroid responsive' limbic encephalitis were caused by one of these autoantibodies. Unlike the classic paraneoplastic encephalitides, autoimmune encephalitis is less likely to be associated with malignancy and can affect patients of all ages.^{2,3} These syndromes can be severe, but respond to immunosuppressive treatments, often with marked recovery. This review aims to provide clinicians with a contemporary overview of the autoimmune encephalitides.

Methods

A literature search was performed using PubMed using the search strategy in Appendix I. Where possible, large studies and review articles were used. Where evidence

did not exist, expert opinion based on our clinical and laboratory experience was utilised.

Anti-NMDA receptor encephalitis

Dalmau and colleagues in 2007 described 12 women with ovarian teratomas who developed prominent psychiatric symptoms, amnesia, seizures, dyskinesia, autonomic dysfunction and a decreased level of consciousness.⁴ These women were shown to have autoantibodies against the NMDA receptor (anti-NMDAR) in both serum and cerebrospinal fluid (CSF).⁴

Following this seminal publication, anti-NMDAR encephalitis has been increasingly recognised with over 700 cases reported by 2014.⁵ Anti-NMDAR encephalitis is now considered to be the most frequently diagnosed of the autoimmune encephalitides and is one of the commonest causes of encephalitis.⁶

Presentation and clinical features

Anti-NMDAR encephalitis has now been defined in several large case series of both adults and children and can present as a paraneoplastic (usually ovarian teratoma) or non-paraneoplastic syndrome.^{5,7-9} The typical presentation is with initial neuropsychiatric features, followed by a severe and prolonged encephalitis^{9,10} (Figure 1). A total of 80% of patients with anti-NMDAR encephalitis are female with a median age of onset of 21 years (age range: 8 months to 85 years).⁵ Within the age groups of less than 12 years or more than 45 years, an increased proportion of male patients (up to 40%) has been reported.⁵

In approximately 70% of patients, there is a prodrome of headache, fever, vomiting, diarrhoea and upper respiratory tract symptoms.^{6,7} It is unclear whether a preceding viral infection precipitates the pathogenesis of the disease or whether this prodrome represents an intrinsic part of the syndrome.

Typically within 2 weeks from onset, prominent psychiatric symptoms and personality changes emerge.^{6,7,10} Common manifestations include anxiety, insomnia, hallucinations, delusions, agitation, mania and hypersexuality, but social withdrawal and stereotypical behaviours may develop.^{6,7} Consequently, the majority of patients in this phase present for psychiatric care. Cognitive changes, particularly impaired attention, short term

memory loss and confusion, are always present and increasing severity over time.^{6,7,11} Problems with language are commonly observed, ranging from reduced verbal output to mutism. Many patients frequently experience fluctuations in mental status resembling delirium.

Focal or generalised seizures (and even status epilepticus) may occur early or later in the disease course.⁸ Seizure activity may be refractory to treatment necessitating the use of multiple anti-epileptic agents.¹² However, the frequency and intensity of the seizures tend to decrease as the disease evolves.⁶

Concurrently, a variety of abnormal movements may occur, including choreiform movements and stereotypies.¹³ Formal electroencephalogram (EEG) evaluation is very important to distinguish between seizure activity and movement disorders, to ensure appropriate treatment.

In the later stages of illness, there is a rapid decrease in responsiveness, with alternating agitation and catatonia, and an inability to follow verbal commands.^{7,10} A range of abnormal movements may be manifest: common orolingual dyskinesias; but also limb and trunk choreoathetosis, elaborate movements of the arms and legs, oculogyric crisis, dystonia, rigidity and opisthotonic postures.⁶

Autonomic dysregulation, including thermodynamic instability, cardiac arrhythmias, blood pressure instability and

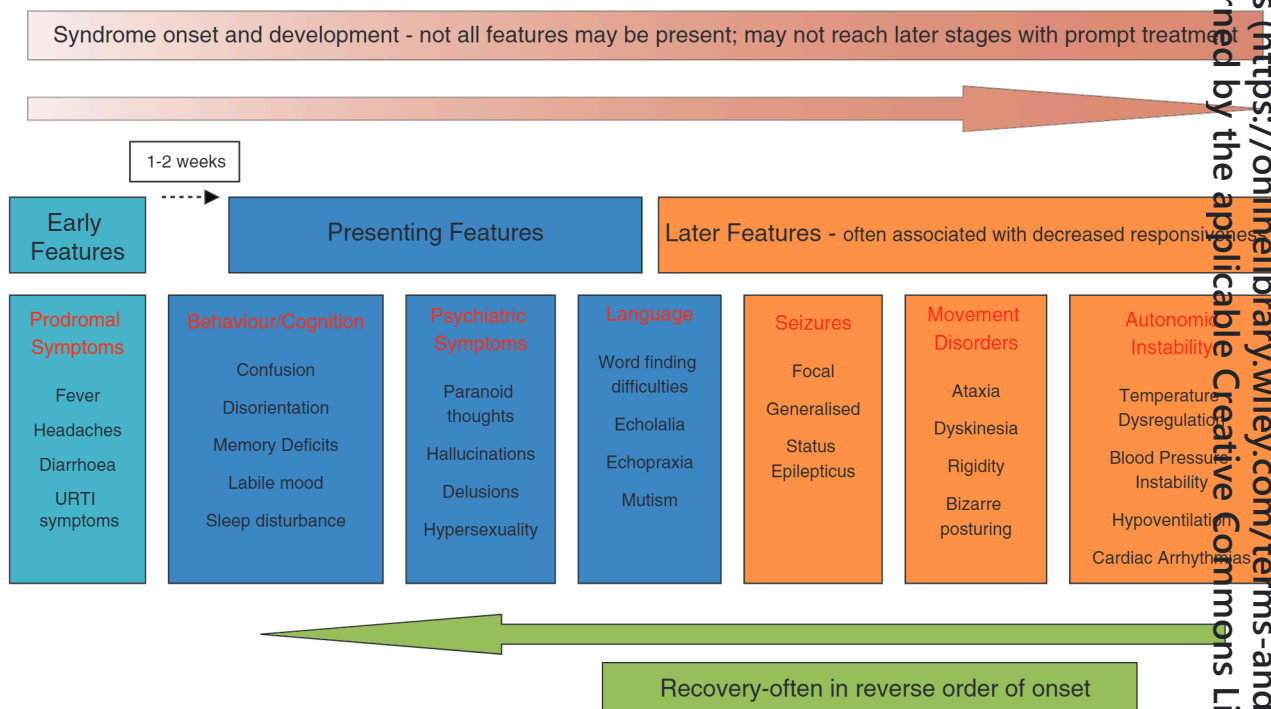


Figure 1 Usual acute clinical presentation of anti-N-methyl-D-aspartic acid receptor (NMDAR) encephalitis through various stages of development

hypersalivation, frequently develop often requiring management in the intensive care setting.^{6,10,11} Hypoventilation may occur as the patient becomes comatose.⁶

While the above descriptions are the most typical and frequently documented, atypical presentations are being increasingly recognised. These include isolated psychiatric manifestations, dystonia or epilepsy syndromes. Pure monosymptomatic syndromes are uncommon, representing less than 5% of patients.⁷ However, most of these patients appear to develop other symptoms of the condition over time.

Finally, there are patients with longstanding psychiatric, cognitive and personality changes, but without the typical presenting symptoms described above who have been found to be anti-NMDAR antibody positive.¹⁴ The variety of initial clinical presentations described above may well lead to the concept of anti-NMDAR spectrum disorders.

Demographics and tumour association

The incidence of anti-NMDAR encephalitis is unknown and likely underestimated in the existing literature due to under-recognition. For example, anti-NMDAR encephalitis was somewhat surprisingly found to be the leading cause of encephalitis in retrospectively tested sera that had been collected by the California Encephalitis Project. In this cohort of patients aged under 30 years, anti-NMDAR encephalitis was four times more common than encephalitis caused by herpes simplex type-1, varicella-zoster virus or West Nile virus and was slightly more common than enterovirus-related encephalitis.¹⁵ However, the authors did raise the possibility that referral bias may have affected these findings, and the discussion did not clarify if other aetiologies had been identified.

In another retrospective study of encephalitis of unknown origin, in patients aged 18–35 years at a single institution intensive care unit over a 5-year period, six out of seven otherwise unknown origin cases were found to be positive for anti-NMDAR antibodies on stored serum/CSF.¹⁶ Anti-NMDAR encephalitis represented 1% of all presentations within this age group in this institution's intensive care unit.¹⁶ A United Kingdom multi-centre population-based study of causes of encephalitis determined that 4% of patients had anti-NMDAR encephalitis, the second most common immune-mediated encephalitis after acute disseminated encephalomyelitis.¹⁷

Teratomas are found in up to one third of adult patients, with females of reproductive age being most commonly affected. In contrast, teratomas are unlikely to be found in children and males.^{4,7,8} Other associated tumours are rare, but include extra-ovarian teratoma,

testicular tumour (including teratoma), neuroblastoma, Hodgkin lymphoma and various tumours of the breast, lung and thymus.^{5,10}

Prognosis

Death from anti-NMDAR encephalitis is uncommon if treatment is prompt and aggressive.⁵ The long-term outlook of an individual patient is dependent on early diagnosis, appropriate treatment and complete removal of the associated tumour in paraneoplastic cases.^{5,18}

Intriguingly, recovery may occur in a multistage process which is often in the reverse order of symptom presentation⁶ (Figure 1). Many patients require hospitalisation for prolonged periods, followed by months of physical and behavioural rehabilitation.^{5,6} Recovery is usually slow and can take up to 18 months. In a cohort of 360 patients, 75% experienced a complete or near complete recovery (with modified Rankin Scale (mRS) of 1–2).⁶ The remainder who survived had sequelae that severely affected their quality of life (mRS > 3). In those who obtained full or near full recovery, 85% still had significant neuropsychiatric symptoms at time of discharge, and in some cases, these deficits were persistent.^{7,19} These included memory deficits and impairment in executive function, with behavioural, language and social skill deficits tending to be the last to resolve.^{7,19} This highlights the importance of a multidisciplinary approach to treatment and rehabilitation.

Encephalitis associated with antibodies to the VGKC complex

Antibodies to the VGKC complex are the second most common cause of autoimmune encephalitis and are associated with a range of clinical syndromes. These include limbic encephalitis, neuromyotonia, Morvan syndrome as well as epilepsy syndromes.

Recently, two distinct major antigens have been identified that are closely related to, but not part of the VGKC: leucine-rich glioma-inactivated 1 (LGI1) and contactin-associated protein-like 2 (CASPR2). The identification of antibodies to these proteins was significant because the majority of high-titre VGKC antibodies are not directed against the VGKC directly, but rather LGI1 or CASPR2 antigens, which result in two distinct clinical syndromes.^{20–22} Many of the cases reported before 2010 with limbic encephalitis and VGKC antibodies were likely related to LGI1 antibodies.²²

Anti-LGI1-associated encephalitis

High-titre anti-VGKC antibodies are most commonly directed against LGI1. Patients with anti-LGI1 antibodies

develop classic symptoms of limbic encephalitis including memory disturbance, confusion, neuropsychiatric features and seizures, including non-convulsive status epilepticus.^{20,21,23} A significant proportion develop hyponatraemia and experience rapid eye movement sleep disturbance.²⁰

Prior to developing the classic symptoms of limbic encephalitis, many patients experience frequent but brief faciobrachial dystonic seizures usually affecting the arm and ipsilateral face.²⁴ Only a proportion of these patients have classical epileptiform changes on EEG. The seizures are often refractory to anti-epileptic drugs, but respond to immunosuppressive/immunomodulatory therapies.²⁵ Limbic encephalitis and temporal lobe seizures can develop later in the disease course. Importantly, anti-LGII-associated limbic encephalitis is usually non-paraneoplastic.^{20,21}

Case series indicate a male preponderance with an average age of 60 years.^{20,21} Anti-LGII-associated limbic encephalitis is reported to generally respond well to promptly administered immunosuppressive regimens, but the long-term outlook is uncertain.^{3,20,21} Some authors report that while the response to treatment is faster, long-term recovery is not as complete as in patients with anti-NMDAR encephalitis,²⁶ although no direct comparison studies exist.

Anti-CASPR2-associated encephalitis

The clinical spectrum associated with anti-CASPR2 antibodies differs from that associated with anti-LGII antibodies. Limbic encephalitis is less frequently observed, whereas peripheral nerve hyperexcitability syndromes, such as Morvan syndrome are more common.^{20,27} Morvan syndrome is a complex disease which combines neuromyotonia with autonomic disturbance and encephalopathy.²⁷ Neuromyotonia associated with anti-CASPR2 antibodies is frequently associated with a painful peripheral neuropathy, but has been rarely reported in isolation.²⁸ Patients can present with bulbar weakness.^{27,28} Muscle atrophy and fasciculations may occur, making anti-CASPR2 an important treatable differential diagnosis of motor neuron disease.²⁸

Positive anti-VGKC not associated with anti-LGII or anti-CASPR2 antibodies

A high titre of anti-VGKC antibodies detected by radio-immunoassay is likely to be clinically relevant and associated with LGII or CASPR2. However, a positive low-titre result not directed against LGII or CASPR2 is frequently encountered and can cause a clinical dilemma.²³

In a recent retrospective study, low positive anti-VGKC values not associated with LGII or CASPR2 were found in patients with neurodegenerative diseases and patients with neurological symptoms, such as myoclonus or dystonia. Low positive anti-VGKC antibodies are likely to be associated with the development of limbic encephalitis.²³

These findings may be due to the presence of antibodies directed against other antigens in the VGKC complex which may be secondary to other pathology or alternatively part of a normal repertoire of autoantibodies found in otherwise healthy individuals which rise to detectable levels under certain circumstances. However, a relatively high proportion of patients (10% of cases) with low-level antibodies have been found to have tumours (breast and endometrial). It remains unclear whether such low positive results necessitate screening for associated malignancies.²³

Anti-GABA_b receptor encephalitis

Anti-gamma-aminobutyric acid_b (anti-GABA_b) receptor antibodies have been reported in a small proportion (5%) of patients with limbic encephalitis.^{29,30} Clinical features consist of memory and behavioural changes in addition to prominent seizures (including status epilepticus).³⁰ Additional presentations include ataxia and opsoclonus-myoclonus.³⁰

In terms of associated malignancies, 50% of patients in a small case series (20 patients) were found to have small-cell lung cancer (SCLC) with a median age of 67 years, whereas patients without SCLC were younger (median age 39 years).³⁰ Males and females appear to be equally affected.²⁹ Anti-GABA_b receptor encephalitis is responsive to immunotherapy, although long-term prognosis is considered to be dictated by any underlying malignancy.³⁰

Anti-GABA_a receptor encephalitis

Anti-gamma-aminobutyric acid_a (anti-GABA_a) receptor encephalitis was first reported in 2014 in six patients (two male children, one female teenager and three adults).³¹ The encephalopathy was acute and progressive with refractory seizures (including status epilepticus) being the predominant factor.³¹ Some patients required pharmacologically induced coma. Immunotherapy led to a partial or complete recovery in the majority of cases, although two patients died due to persistent status epilepticus or sepsis.³¹ Interestingly, other autoimmune diseases or associated antibodies were detected, primarily anti-thyroid peroxidase antibodies and anti-glutamic acid decarboxylase 65 (anti-GAD65) antibodies in other

intracellular antigens which themselves have been associated with encephalopathy).³¹ Three of these six patients would have fulfilled the diagnostic criteria for steroid-responsive encephalopathy related to autoimmune thyroiditis (Hashimoto encephalopathy).³¹ This highlights the importance of testing for anti-GABA_A receptor antibodies when considering a diagnosis of Hashimoto encephalopathy or anti-GAD65 encephalitis.

Anti-AMPA receptor encephalitis

Encephalitis due to the presence of α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (anti-AMPA) receptor antibodies was first described in 2009.³² These patients presented with symptoms typical of a rapidly progressive limbic encephalitis,^{32,33} although acute psychosis has been reported.³⁴ Patients were usually middle aged with a strong female preponderance.^{32,33} The majority of cases reported thus far have been found to have an underlying tumour (including breast, lung and thymus).^{32,33} While the syndrome is responsive to immunotherapy and may be totally reversible, it has been reported to have a tendency to relapse, sometimes repeatedly, despite removal of the tumour.³²

Other autoimmune encephalitis syndromes associated with antibodies against neuronal cell surface and synaptic antigens

Several other autoantibodies associated with encephalitis have been reported, although they remain rare (Table 1). Testing for these antibodies remains largely at the research level.

Antibody testing in autoimmune encephalitis

Notwithstanding the fact that there are no standardised, consensus-agreed clinical criteria for anti-NMDAR encephalitis, all patients with anti-NMDAR encephalitis have antibodies detectable in CSF, however, in 14% of patients antibodies in the serum are not found.^{26,35} In our laboratory, this percentage may be as high as 30% (unpublished data). This is likely due to the fact that CSF is tested undiluted in most laboratories (including ours), compared with 1:10–1:40 dilution for serum specimens. Thus, it is critical that CSF analysis be carried out when anti-NMDAR encephalitis is suspected and no antibody can be detected in the serum. CSF testing is preferred for anti-AMPA and anti-GABA_B receptor antibodies.²⁶ On the other hand, serum is the specimen of choice for the

detection of antibodies against LGI1, CASPR2 and GABA_A receptor.^{26,36} Therefore, both serum and CSF should be sent for testing in patients with suspected autoimmune encephalitis.

The reliability of serum antibody titres as a biomarker of disease activity remains controversial. In anti-NMDAR encephalitis, antibody titres in CSF, and to a lesser extent in serum, correlate with clinical outcome.³⁵ CSF antibody titres associate better with the disease course and relapses than serum titres.³⁵ However, anti-NMDAR antibody titres should not be used solely to guide treatment decisions due to technical reproducibility of testing and long-term persistence of antibodies even after clinical recovery.²⁶

At Pathology Queensland, anti-NMDAR antibody titres are not routinely performed as the commercial NMDAR NR1 subunit-transfected cell lines used by Australian diagnostic laboratories are not readily amenable for end-point titration.³⁷ The alternative approach developed in our laboratory involves estimation of fluorescent intensity of staining using concurrent testing of the latest specimen and the original (positive) sample. The immunofluorescence intensity of the transfected cells on both samples is then compared visually using digital photographs on the same digital platform, any discernible changes in intensity are reported.³⁷

Suspected autoimmune encephalitis with no autoimmune encephalitis autoantibody detected

In patients clinically suspected to have autoimmune encephalitis, but with no detectable autoantibody in serum or CSF, recollection and repeat testing of both serum and CSF is recommended.³⁸ This is especially important if only serum was initially tested.²²

It is important to note that treatment with intravenous immunoglobulin (IVIG) prior to the collection of serum can make the interpretation of anti-NMDAR transfected slides in the laboratory more difficult and potentially lead to false negative results. However, the prior use of IVIG does not affect the interpretation of CSF specimens. Therefore, liaison with the testing laboratory and supervising immunopathologist should be contacted regarding interpretation of testing in this scenario.

Once other causes of encephalitis have been excluded and the clinical scenario remains suggestive of autoimmune encephalitis, a trial of immunosuppression with corticosteroids and IVIG +/- plasmapheresis should be considered and the clinical response carefully observed.²²

Treatment of autoimmune encephalopathies

Treatment regimens used in autoimmune encephalitis are based on the principles of treatment of other life-threatening autoimmune diseases. Various therapies, including corticosteroids, IVIG, plasma exchange, rituximab and cyclophosphamide are currently used (Figure 2). However, there is no clear consensus and no controlled randomised clinical trial data regarding which immunotherapies should be used, the order in which they should be applied or the duration of therapy. Most experience has been attained in the treatment of the two most common diseases – anti-NMDR and anti-LGI1 encephalitis.

In a large multi-institutional observational cohort study of 577 patients with anti-NMDAR encephalitis, 92% of patients received first-line immunotherapy, including steroids, IVIG and/or plasma exchange, combined with tumour removal (when applicable). Improvement occurred within 4 weeks in 53% of patients. A total of 97% demonstrated a good outcome at 24 months as defined by mRS 0-2.⁵ The remaining 47% of patients who did not respond to first-line treatment, generally had a poorer prognosis.⁵ However, non-responders who subsequently received rituximab and/or cyclophosphamide had better outcomes than those who continued on first-line treatments or did not receive any further immunotherapy.⁵

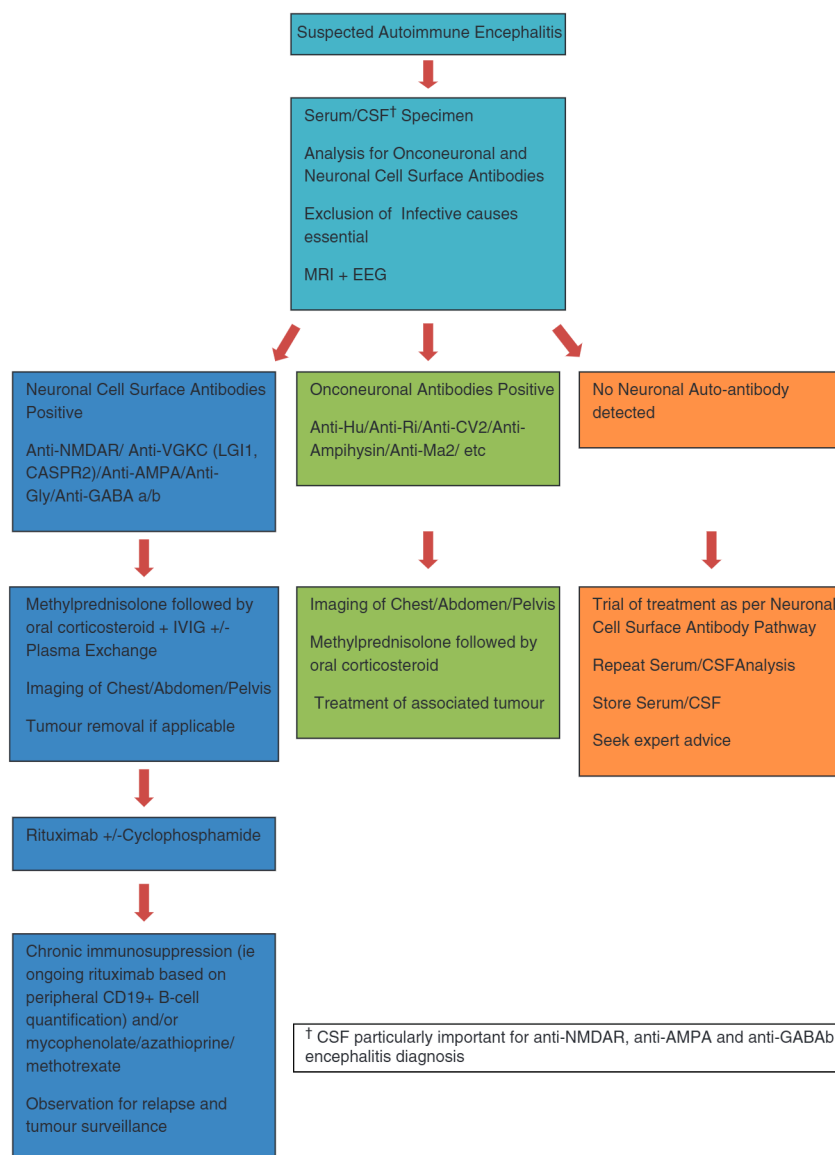


Figure 2 Treatment algorithm for suspected autoimmune encephalitis.

In several autoimmune encephalitides, including anti-NMDAR encephalitis, there is evidence of intrathecal synthesis of antibody.¹⁰ Therefore, therapies which primarily reduce the amount of antibody in the circulation, rather than the CNS, such as IVIG or plasmapheresis, may have only limited efficacy.²²

Relapses in anti-NMDAR encephalitis patients with non-paraneoplastic-associated disease have been reported to occur in around 10–25% of patients.^{9,39} Relapses may be separated by periods of months to years and occur more frequently in patients who are undertreated or not treated with immunotherapy during the first episode of disease.^{9,39}

In the largest study to date, those treated with second-line therapies, such as rituximab and/or cyclophosphamide, were less likely to relapse than those who only received first-line immunosuppressive treatment.⁵ The prolonged use of other immunosuppressive drugs, such as mycophenolate or azathioprine, may decrease the risk of recurrence.⁶ Relapses are rarer in paraneoplastic cases, where symptoms usually improve after complete tumour removal.⁵

As anti-LGI1 encephalitis is less common than anti-NMDAR encephalitis, only smaller published series are available. Patients with anti-LGI1 encephalitis often respond more quickly to IVIG, corticosteroids and plasma exchange than those with anti-NMDAR encephalitis.²⁶ This may be due to the greater production of pathogenic anti-LGI1 antibodies outside of the CNS compared with anti-NMDAR antibodies.²⁶ A total of 80% of patients is reported to have a good outcome, but the mortality rate is around 6%, and approximately 15% of cases will relapse.^{20,21}

In our experience, early treatment with rituximab and/or cyclophosphamide is beneficial in both paraneoplastic and non-paraneoplastic anti-NMDAR antibody-mediated encephalitis. Onset of action for some immunosuppressive agents may take several weeks (such as with rituximab which takes 2–3 weeks for peripheral B-cell depletion to be complete) and therefore, valuable treatment time may be lost whilst waiting for response to first-line agents (corticosteroids, IVIG and/or plasmapheresis). Clinical improvement, even after aggressive therapy with rituximab or cyclophosphamide, is often slow with average acute hospital admission duration reported to be around 3 months.²⁶

References

- 1 Corsellis JA, Goldberg GJ, Norton AR. "Limbic encephalitis" and its association with carcinoma. *Brain* 1968; **91**: 481–96.
- 2 Graus F, Saiz A, Dalmau J. Antibodies and neuronal autoimmune disorders of the CNS. *J Neurol* 2010; **257**: 509–17.
- 3 Leypoldt F, Wandinger KP, Bien CG, Dalmau J. Autoimmune encephalitis. *Eur Neurol Rev* 2013; **8**: 31–7.
- 4 Dalmau J, Tuzun E, Wu HY, Masjuan Rossi JE, Voloschin A *et al.*

Screening and surveillance for underlying tumours

Some authors have recommended that all patients with autoimmune encephalitis should be screened for tumours at the onset of disease, but others have argued that tumour screening is less important in patients with antibodies, such as anti-LGI1 and anti-GABA_A, which have a low association with underlying tumours. If the initial tumour screening is negative in patients with antibodies, such as anti-NMDAR in young adult women, anti-CASPR2, anti-AMPA receptor and anti-GABA_A receptor with a high reported association with underlying tumours, repeat screening at 3–6 months followed by screening at 6 monthly intervals for 4 years is recommended.⁴⁰

The modality of screening will be determined by the age and sex of the patient as well as the particular autoantibody detected. For example, to screen for ovarian teratoma, transvaginal ultrasound followed by computed tomography scan or magnetic resonance imaging of the pelvis is recommended, and the reader is referred elsewhere for discussion of strategies of screening for other malignancies.⁴⁰

Conclusion

The autoimmune encephalitides associated with surface receptors are an important and rapidly emerging group of antibody-associated neurological diseases. Recognition of these conditions is crucial as with prompt diagnosis and treatment the majority have favourable outcomes. Undoubtedly, additional autoantibodies will continue to be detected and the pathogenic mechanisms will continue to be elucidated. Close liaison with the testing laboratory and immunopathologist is essential.

Rapid treatment with immunotherapy and tumour removal, if applicable, remains crucial to prevent progressive neurological decline and leads to considerable clinical improvement in most cases. Patients need to be followed closely after recovery for signs of relapse. Acute and long-term treatment regimens will continue to evolve, with treatment regimens in the rarer syndromes likely to be guided by those developed for anti-NMDAR encephalitis.

- Paraneoplastic anti-*N*-methyl-D-aspartate receptor encephalitis associated with ovarian teratoma. *Ann Neurol* 2007; **61**: 25–36.
- 5 Titulaer MJ, McCracken L, Gabilondo I, Armangue T, Glaser C, Iizuka T *et al.* Treatment and prognostic factors for long-term outcome in patients with anti-NMDA receptor encephalitis: an observational cohort study. *Lancet Neurol* 2013; **12**: 157–65.
 - 6 Dalmau J, Lancaster E, Martinez-Hernandez E, Rosenfeld MR, Balice-Gordon R. Clinical experience and laboratory investigations in patients with anti-NMDAR encephalitis. *Lancet Neurol* 2011; **10**: 63–74.
 - 7 Dalmau J, Gleichman AJ, Hughes EG, Rossi JE, Peng X, Lai M *et al.* Anti-NMDA-receptor encephalitis: case series and analysis of the effects of antibodies. *Lancet Neurol* 2008; **7**: 1091–8.
 - 8 Florance NR, Davis RL, Lam C, Szperka C, Zhou L, Ahmad S *et al.* Anti-*N*-methyl-D-aspartate receptor (NMDAR) encephalitis in children and adolescents. *Ann Neurol* 2009; **66**: 11–18.
 - 9 Irani SR, Bera K, Waters P, Zuliani L, Maxwell S, Zandi MS *et al.* *N*-methyl-D-aspartate antibody encephalitis: temporal progression of clinical and paraclinical observations in a predominantly non-paraneoplastic disorder of both sexes. *Brain* 2010; **133**: 1655–67.
 - 10 Vincent A, Bien CG, Irani SR, Waters P. Autoantibodies associated with diseases of the CNS: new developments and future challenges. *Lancet Neurol* 2011; **10**: 759–72.
 - 11 Peery HE, Day GS, Dunn S, Fritzler MJ, Pruss H, De Souza C *et al.* Anti-NMDA receptor encephalitis. The disorder, the diagnosis and the immunobiology. *Autoimmun Rev* 2012; **11**: 863–72.
 - 12 Day GS, High SM, Cot B, Tang-Wai DF. Anti-NMDA-receptor encephalitis: case report and literature review of an under-recognized condition. *J Gen Intern Med* 2011; **26**: 811–16.
 - 13 Luca N, Daengsuwan T, Dalmau J, Jones K, deVeber G, Kobayashi J *et al.* Anti-*N*-methyl-D-aspartate receptor encephalitis: a newly recognized inflammatory brain disease in children. *Arthritis Rheum* 2011; **63**: 2516–22.
 - 14 Kayser MS, Titulaer MJ, Gresa-Arribas N, Dalmau J. Frequency and characteristics of isolated psychiatric episodes in anti-*N*-methyl-D-aspartate receptor encephalitis. *JAMA Neurol* 2013; **70**: 1133–9.
 - 15 Gable MS, Sheriff H, Dalmau J, Tilley DH, Glaser CA. The frequency of autoimmune *N*-methyl-D-aspartate receptor encephalitis surpasses that of individual viral etiologies in young individuals enrolled in the California Encephalitis Project. *Clin Infect Dis* 2012; **54**: 899–904.
 - 16 Pruss H, Dalmau J, Harms L, Holtje M, Ahnert-Hilger G, Borowski K *et al.* Retrospective analysis of NMDA receptor antibodies in encephalitis of unknown origin. *Neurology* 2010; **75**: 1735–9.
 - 17 Granerod J, Ambrose HE, Davies NW, Clewley JP, Walsh AL, Morgan D *et al.* Causes of encephalitis and differences in their clinical presentations in England: a multicentre, population-based prospective study. *Lancet Infect Dis* 2010; **10**: 835–44.
 - 18 Wandinger KP, Saschenbrecker S, Stoecker W, Dalmau J. Anti-NMDA-receptor encephalitis: a severe, multistage, treatable disorder presenting with psychosis. *J Neuroimmunol* 2011; **231**: 86–91.
 - 19 Kayser MS, Kohler CG, Dalmau J. Psychiatric manifestations of paraneoplastic disorders. *Am J Psychiatry* 2010; **167**: 1039–50.
 - 20 Irani SR, Alexander S, Waters P, Kleopa KA, Pettingill P, Zuliani L *et al.* Antibodies to Kv1 potassium channel-complex proteins leucine-rich, glioma inactivated 1 protein and contactin-associated protein-2 in limbic encephalitis, Morvan's syndrome and acquired neuromyotonia. *Brain* 2010; **133**: 2734–48.
 - 21 Lai M, Huijbers MG, Lancaster E, Graus F, Bataller L, Balice-Gordon R *et al.* Investigation of LGI1 as the antigen in limbic encephalitis previously attributed to potassium channels: a case series. *Lancet Neurol* 2010; **9**: 776–85.
 - 22 Linnoila JJ, Rosenfeld MR, Dalmau J. Neuronal surface antibody-mediated autoimmune encephalitis. *Semin Neurol* 2014; **34**: 458–66.
 - 23 Paterson RW, Zandi MS, Armstrong R, Vincent A, Schott JM. Clinical relevance of positive voltage-gated potassium channel (VGKC)-complex antibodies: experience from a tertiary referral centre. *J Neurol Neurosurg Psychiatry* 2014; **85**: 625–30.
 - 24 Irani SR, Michell AW, Lang B, Pettingill P, Waters P, Johnson MR *et al.* Faciobrachial dystonic seizures precede Lgi1 antibody limbic encephalitis. *Ann Neurol* 2011; **69**: 892–900.
 - 25 Irani SR, Stagg CJ, Schott JM, Rosenthal CR, Schneider SA, Pettingill P *et al.* Faciobrachial dystonic seizures: the influence of immunotherapy on seizure control and prevention of cognitive impairment in a broadening phenotype. *Brain* 2013; **136**: 3151–62.
 - 26 Leypoldt F, Armangue T, Dalmau J. Autoimmune encephalopathies. *Ann NY Acad Sci* 2015; **1338**: 94–114.
 - 27 Irani SR, Pettingill P, Kleopa KA, Schiza N, Waters P, Mazia C *et al.* Morvan syndrome: clinical and serological observations in 29 cases. *Ann Neurol* 2012; **72**: 241–55.
 - 28 Lancaster E, Huijbers MG, Bar V, Boronat A, Wong A, Martinez-Hernandez E *et al.* Investigations of caspr2, an autoantigen of encephalitis and neuromyotonia. *Ann Neurol* 2011; **69**: 303–11.
 - 29 Lancaster E, Martinez-Hernandez E, Dalmau J. Encephalitis and antibodies to synaptic and neuronal cell surface proteins. *Neurology* 2011; **77**: 179–89.
 - 30 Hofberger R, Titulaer MJ, Sabater A, Dome B, Rozsas A, Hegedus B *et al.* Encephalitis and GABAB receptor antibodies: novel findings in a new case series of 20 patients. *Neurology* 2011; **78**: 1500–6.
 - 31 Petit-Pedrol M, Armangue T, Peng X, Bataller L, Cellucci T, Davis R *et al.* Encephalitis with refractory seizures, status epilepticus, and antibodies to the GABAA receptor: a case series, characterisation of the antigen, and analysis of the effects of antibodies. *Lancet Neurol* 2014; **13**: 276–86.
 - 32 Lai M, Hughes EG, Peng X, Zhou L, Gleichman AJ, Shu H *et al.* AMPA receptor antibodies in limbic encephalitis alter synaptic receptor location. *Ann Neurol* 2009; **65**: 424–34.
 - 33 Bataller L, Galiano R, Garcia-Escribana M, Martinez B, Sevilla T, Blasco R *et al.* Reversible paraneoplastic limbic encephalitis associated with antibodies to the AMPA receptor. *Neurology* 2010; **74**: 265–7.
 - 34 Graus F, Boronat A, Xifro X, Boix A, Svigelj V, Garcia A *et al.* The expanding

- clinical profile of anti-AMPA receptor encephalitis. *Neurology* 2010; **74**: 857–9.
- 35 Gresa-Arribas N, Titulaer MJ, Torrents A, Aguilar E, McCracken L, Leypoldt F *et al.* Antibody titres at diagnosis and during follow-up of anti-NMDA receptor encephalitis: a retrospective study. *Lancet Neurol* 2014; **13**: 167–77.
- 36 Sinmaz N, Amatoory M, Merheb V, Ramanathan S, Dale RC, Brilot F. Autoantibodies in movement and psychiatric disorders: updated concepts in detection methods, pathogenicity, and CNS entry. *Ann N Y Acad Sci* 2015; **1351**: 22–38.
- 37 Wilson BP, Prain K, Gillis D, Wong R. Laboratory testing for anti-NMDAR in autoimmune encephalitis: the HSSA-pathology Queensland experience. *Pathologist* 2013; **1**: 7–12. Available from URL: WWW.hkcpth.org.
- 38 Lancaster E, Dalmau J. Neuronal autoantigens – pathogenesis, associated disorders and antibody testing. *Nat Rev Neurol* 2012; **8**: 380–90.
- 39 Gabilondo I, Saiz A, Galan L, Gonzalez V, Jadraque R, Sabater L *et al.* Analysis of relapses in anti-NMDAR encephalitis. *Neurology* 2011; **77**: 996–999.
- 40 Titulaer MJ, Sofiotti R, Dalmau J, Gilhus NE, Giometto B, Graus F *et al.* Screening for tumours in paraneoplastic syndromes: report of an EFNS task force. *Eur J Neurol* 2011; **18**: 19–e3.
- 41 Carvajal-Gonzalez A, Leite MI, Waters P, Woodhall M, Coutinho E, Balint B *et al.* Glycine receptor antibodies in PERM and related syndromes: characteristics, clinical features and outcomes. *Brain* 2014; **137**: 2178–92.
- 42 Lancaster E, Martinez-Hernandez E, Titulaer MJ, Boulos M, Weaver S, Antoine JC *et al.* Antibodies to metabotropic glutamate receptor 5 in the Ophelia syndrome. *Neurology* 2011; **77**: 1698–701.
- 43 Mat A, Adler H, Merwick A, Chadwick G, Gullo G, Dalmau JO *et al.* Ophelia syndrome with metabotropic glutamate receptor 5 antibodies in CSF. *Neurology* 2013; **80**: 1349–50.
- 44 Balint B, Jarius S, Nagel S, Haberkorn U, Probst C, Blocker *et al.* Progressive encephalomyelitis with rigidity and myoclonus: a new variant with DPPX antibodies. *Neurology* 2014; **82**: 1521–8.
- 45 Boron A, Gelfand JM, Gresa-Arribas N, Jeong HY, Walsh M, Roberts K *et al.* Encephalitis and antibodies to dipeptidyl-peptidase-like protein-6, a subunit of Kv4.2 potassium channels. *Ann Neurol* 2013; **73**: 120–8.
- 46 Dale RC, Merheb V, Pillai S, Wang D, Cantrill L, Murphy TK *et al.* Antibodies to surface dopamine-2 receptor in autoimmune movement and psychiatric disorders. *Brain* 2012; **135**: 3453–68.

Appendix I

Search strategy for literature on autoimmune encephalitis

Database: PubMed

Search Strategy: (((((encephalitis) OR encephalopath*) AND autoimmune) AND limbic) OR "receptor encephalitis AND english[Language] (653 references)